Generating Long Time Series of High Spatiotemporal Resolution FPAR Images in the Remote Sensing Trend Surface Framework

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***Abstract*— To improve our capacity to map long-term vegetation dynamics in heterogeneous landscapes, this study proposed a new prior knowledge-based spatiotemporal enhancement method, namely, PK-STEM, to fuse MODIS and Landsat FPAR products following the remote sensing trend surface framework. PK-STEM uses historical Landsat FPAR images as prior knowledge and fuses them with new satellite-derived FPAR data. PK-STEM can work in three modes: 1) using only MODIS data; 2) using only Landsat data; and 3) using both MODIS and Landsat data. This study retrieved FPAR from Landsat images using a scaling-based method and tested the performance of PK-STEM in a regional application. For the entire year of 2012, we compared the performance of PK-STEM in different modes and with that of two typical spatiotemporal fusion methods, the enhanced spatial and temporal adaptive reflectance model (ESTARFM) and unmixing-based linear mixing growth model (LMGM). Then, a long time series FPAR data set at 30-m resolution and eight-day intervals was generated for 13 years (2000–2012). Our results show that PK-STEM in mode III is the most robust and accurate (root mean squared error (RMSE) = 0.062; mean R = 0.851) among the three modes and more accurate than ESTARFM (mean RMSE = 0.065; mean R = 0.776) and LMGM (mean RMSE = 0.074; mean R = 0.734). For the 12 years (2000–2011), PK-STEM also achieves high accuracies with mean RMSE = 0.066 and R = 0.938. PK-STEM is very flexible with a continual update mechanism and is efficient for long time series applications.**

***Index Terms*— Fraction of absorbed photosynthetically active radiation (FAPAR/FPAR), spatiotemporal data fusion, Landsat, MODIS, PK-STEM, remote sensing trend surface (RSTS), vegetation dynamic.**

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I. Introduction

VEGETATION sustains life on Earth by synthesizing sugars from water and CO_2 using absorbed photosynthetically active radiation (APAR) and cools the surface by transpiring large amounts of water during this process [1]. The fraction of APAR is defined as FPAR [2], a critical indicator of vegetation growth and energy and carbon exchange between the biosphere and atmosphere. Spatially explicit and temporally continuous FPAR is a key input to numerous land surface process models, such as those used to calculate evapotranspiration and net primary production [3]–[6]. While current global satellite products can provide spatially and temporally continuous FPAR data sets at coarse resolutions (250 m to 1 km), studies of ecological processes and impacts in heterogeneous landscapes require frequent and finer-resolution FPAR data. To differentiate short-term from long-term trends, such fine-scale FPAR estimates must be collected for long time periods [7]. Unfortunately, due to technical limitations and unfavorable imaging conditions, no single sensor can provide long-term satellite observations with a high spatiotemporal resolution [8]. Spatiotemporal fusion of frequent coarse-resolution data and less frequent fine-resolution data has been a feasible solution to synthesize data with a high spatiotemporal resolution [9]. However, despite the many methods developed for this purpose, it is still challenging to generate a long time series (e.g., >5 years) FPAR data set with a high spatiotemporal resolution for continuous vegetation monitoring, such as in the estimation of crop yield and carbon exchange over multiple years.

Current spatiotemporal fusion methods can be categorized into five types [9]: weight function-based [10]–[15], unmixing-based [16]–[19], Bayesian-based [20], learning-based [21]–[23], and hybrid methods that combine two or more techniques from the aforementioned types [8], [14], [15], [24]. All of these methods use one or more pairs of coarse- and fine-resolution images (hereafter referred to as coarse and fine images, respectively) as input base data and predict a fine image with a coarse image at a prediction date.

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While the above methods can restore vegetation dynamics at a fine resolution, sometimes with good results, their major limitations have precluded their use in generating long time series of high spatiotemporal resolution data. First, all of these methods use spatial information from the input fine images to downscale the temporal information from coarse images to subpixel scales [25]. Such a downscaling process is quite uncertain in regard to spatiotemporal continuity with only one or two input fine images. Second, all the current methods use a fixed framework to blend coarse and fine images, e.g., a fixed weighting system in the weight function-based methods. They cannot readily integrate more information, such as that found in historical fine images. It is also impossible for these methods to work with either coarse or fine images in some extreme cases, or to update the generated high spatiotemporal resolution data when new data are available. Third, the current methods are generally time-consuming, and it would cost tremendous amounts of time to calculate the year-round data for many years. For instance, classic weight function-based methods, such as the spatial and temporal adaptive reflectance fusion model (STARFM) [10] and enhanced STARFM (ESTARFM) [12], need to process each pixel in a fine image (hereafter referred to as "fine pixel") with neighborhood calculations. Millions of iterative calculations are thus required to predict a single fine image.

A recent trend in spatiotemporal data fusion is to use Kalman filters to fuse coarse and fine images [26]–[28]. However, all these methods still use limited fine images to downscale the temporal information from coarse pixels to predict values for fine pixels. The downscaling process remains uncertain and can rarely incorporate additional information from historical fine images or newly available data. In particular, the Kalman filter reflectance fusion model [28] solves the reflectance change rate for fine-resolution endmembers from coarse images through a linear mixture model to predict fine pixel values. The Earth Observation Land Data Assimilation System (EO-LDAS), a scheme designed to improve the mapping of biophysical parameters in a weak constraint variational data assimilation system, has the potential to fuse multiresolution data, yet its current version only implemented a linear process model [29], [30].

Therefore, this study proposes a prior knowledge-based spatiotemporal enhancement method, namely, PK-STEM, for fusing satellite-derived FPAR products with different resolutions following the remote sensing trend surface (RSTS) conceptual framework. Academician Xiaowen Li initially proposed the RSTS conceptual framework and defined an RSTS as a scale-consistent remote sensing product with a high spatiotemporal resolution [31], [32]. The RSTS framework provides a complete view of the scaling issues in remote sensing and has inspired a series of related studies, such as validating coarse-resolution satellite products via *in situ* measurements [33], [34] and rapid retrieval of FPAR from fine-resolution satellite data [35]. The RSTS framework prioritizes the use and update of prior knowledge with new satellite observations in a flexible framework. Although it is conceptual and has many technical issues that need to be detailed, the RSTS framework offers a new perspective for spatiotemporal data fusion.

Within the RSTS framework, PK-STEM uses historical fine images as a source of additional information and integrates a series of techniques to fuse spatiotemporal information from coarse and fine FPAR images.

We demonstrate the steps of PK-STEM in blending MODIS and Landsat FPAR data sets to synthesize a new FPAR data set at 30-m resolution and eight-day intervals, in which we initially retrieved FPAR from Landsat reflectance through a scaling-based method [35]. For the entire year in 2012, we first compare the performance of PK-STEM using different input data, i.e., only MODIS data, only Landsat data, or both MODIS and Landsat data. Then, we compare the PK-STEM using both MODIS and Landsat data with two typical spatiotemporal data fusion methods. The weight function-based ESTARFM [12] and unmixing-based linear mixing growth model (LMGM) [36] are selected because they are representative of the two types of methods, have high accuracy and are easy to implement. Then, PK-STEM is applied to generate a 13-year FPAR data set at 30-m resolution and eight-day intervals.

II. Prior Knowledge-Based Methodology for Long-Time-Series Spatiotemporal Enhancement

A. Algorithm Development in the RSTS Framework

Drawing inspiration from the idea of a trend surface in geography, Xiaowen Li defines an RSTS as a set of scale-consistent prior knowledge data with a high spatiotemporal resolution [31], [32]. The particular features of an RSTS include the following: 1) it provides continuous spatial and temporal variations of the land surface through a high spatiotemporal resolution; 2) its derivation prioritizes the use of prior knowledge, and it can also serve as prior knowledge for further updates; and 3) it is generated in a scale-consistent manner to guarantee flux modeling consistency at different spatial resolutions.

A trend surface in geography represents the large-scale spatial variation [37] and is generally described by a 2-D polynomial equation of the first, second, or higher degree [38]. In statistics, 1-D time series data temporally consist of trend, seasonal, and remainder components, corresponding to the low frequency variation with nonstationary and long-term changes, seasonal frequency variation, and the remaining variation in the data [39]. Similar to that of a trend surface in geography, the idea of an RSTS is to separate time series of observable surfaces in some year into large- and small-scale variations, which correspond to trend and residual surfaces, respectively [31], [32]. Trend surfaces are very similar to the actual surfaces in terms of large-scale spatial variations and seasonal variations, which can be obtained from time series data in some year. Residual surfaces represent the differences between the trend and actual surfaces, which can be corrected by fusing new observations.

Since no single sensor can provide satellite data with a high spatial resolution (i.e., no coarser than 30 m) and a high temporal frequency (i.e., no longer than eight-day intervals) for a long term (e.g., more than ten years), Li proposed constructing prior RSTSs from prior knowledge in terms of

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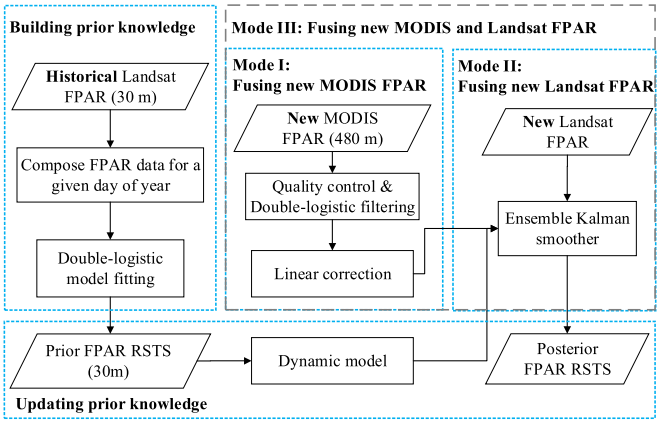


Fig. 1. Flowchart of PK-STEM for spatiotemporal fusion of satellite-derived FPAR with different resolutions (RSTS = remote sensing trend surface).

accumulated data, models, and related geoscience knowledge and then fusing these prior trend surface images with new satellite data through Bayes' theorem. The output is a series of posterior trend surface images that cover all of our knowledge about the spatiotemporal variations of the target parameter and can serve as prior knowledge for further studies [31], [32]. He integrated all the potential information sources and necessary steps into one conceptual framework for the generation of an RSTS data set.

To meet the scale-consistent requirement within the RSTS concept, we first estimated FPAR from fine images through scale-consistent downscaling of coarse FPAR products. We developed PK-STEM to fuse FPAR data sets with different resolutions. Within the RSTS framework, PK-STEM utilizes historical fine FPAR images to construct prior FPAR trend surface images with a high spatiotemporal resolution. PK-STEM then fuses these prior RSTS images with newly observed coarse and fine FPAR images through linear correction and an ensemble Kalman smoother (EnKS), respectively. The present implementation specifically deals with, yet is not limited to, the blending of MODIS FPAR products and Landsat FPAR images. Fig. 1 presents the flowchart of PK-STEM for fusing satellite-derived FPAR data sets with different resolutions.

PK-STEM provides a very flexible framework for spatiotemporal data fusion. It can run in three modes: 1) using only MODIS FPAR data (mode I); 2) using only Landsat FPAR data (mode II); and 3) using both MODIS and Landsat FPAR data (mode III).

B. FPAR Estimation at a Fine Resolution

We employed a scaling-based method to estimate FPAR from Landsat images using MODIS FPAR products as a reference [35]. To avoid scale effects in the scaling process, the method models the pixel-level FPAR as a linear combination of multiband reflectance, that is, reflectance in bands 2 to 5 (i.e., green, red, near-infrared, and shortwave) from Landsat TM/ETM+ data. The method assumes a date- and land cover-specific linear FPAR retrieval model to represent the influences of time-varying and spatially heterogeneous factors on the complex relationship between FPAR and reflectance. The scaling-based method thus builds the linear FPAR retrieval model through multivariant linear regression analysis between high-quality FPAR retrievals from homogeneous MODIS pixels and the aggregated Landsat reflectance for each land cover type on the same date. More details of the method can be found in [35].

The scaling-based method can provide reliable MODIS-consistent FPAR estimations at the Landsat resolution by inheriting the best-quality FPAR retrievals from the MODIS products. It is simple and transferable through space and time, and can thus be automated for regional and long-term applications.

C. Composing Prior FPAR Trend Surface Images

PK-STEM composes multiyear historical Landsat FPAR images for a given day of the year (DOY) to generate a frequent but irregularly temporally spaced FPAR data set. This composite data set probably has inconsistent temporal curves due to interannual variations and quality decline in the available images. For landscapes with single-season variation, the basis form of a double logistic model [40] is used to filter the composite FPAR data set and is expressed as

$$\mathrm{FPAR}_t = v_b + \frac{k}{1+\exp(-c(t-p))} - \frac{k+v_b-v_e}{1+\exp(-d(t-q))} \tag{1}$$

where FPAR_t is the FPAR value for a given DOY t; k represents the maximum value of FPAR; c and d represent the first (located in summer) and second (located in winter) inflection points of the crop growth curve, respectively; p and q represent the dates corresponding to c and d, respectively; and v_b and v_e are the FPAR values at the starting date and ending date of crop growth, respectively.

To avoid the influence of data gaps and clouds, missing or low-quality FPAR values are excluded from the fitting process. The ordinary least squares method is employed to retrieve the parameters in (1) for each Landsat pixel within the composite FPAR data set. The root mean squared error (RMSE) of the residuals is used as an indicator of the uncertainty of the curve fitting for each Landsat pixel.

With the use of the fit double logistic model, we can compute a new FPAR data set at 30-m resolution and eight-day intervals (i.e., the same interval as that of MODIS FPAR products), and we will hereafter refer to this new data set as the prior FPAR RSTS data. Derived from multiyear Landsat images, the prior FPAR RSTS data might not capture abrupt land cover changes or phenological changes for a single year but represent the average status of vegetation dynamics over many years with good spatiotemporal continuity at a high spatiotemporal resolution.

D. FPAR Dynamic Model

PK-STEM uses a dynamic model to represent the spatiotemporal information in the prior FPAR RSTS data. The dynamic model describes the change in FPAR with time through the

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following equations:

$$X_t = M_t X_{t-1} \tag{2}$$

$$M_t = 1 + \frac{1}{Z_t + \varepsilon} \times \frac{\mathrm{d}Z_t}{\mathrm{d}t} \tag{3}$$

where X_t and X_{t-1} represent the FPAR values at dates t and $t-1$, respectively, and M_t is a linear operator that predicts X_t from X_{t-1}. M_t is calculated from the temporal profile from the prior FPAR RSTS data using (3), where Z_t is the FPAR value from the prior FPAR RSTS data at t and ε is set to 0.001 to prevent a null denominator.

Using the above equations, the dynamic model provides a prior estimate of FPAR for each date (i.e., a prior FPAR image), that will be further updated by new satellite FPAR images (MODIS, Landsat, or both) to incorporate real-time changes.

E. Fusing New Satellite FPAR Data

The prior FPAR RSTS data are generally derived from historical time series data. These RSTS images might be very similar to the actual images in terms of large-scale spatial variations and seasonal variations but can hardly capture small-scale variations (e.g., changes in land cover or vegetation phenology) correctly for a specific year. Thus, they still need to be corrected by fusing new observations to restore the actual spatial and temporal variations in a specific year. In PK-STEM, new satellite data can be fused in three modes as follows.

1) Mode I : Fusing New MODIS FPAR Data: Since Landsat images are always sparse in time, it is desirable to integrate the new frequent information from MODIS FPAR data to adjust the prior FPAR RSTS data. All the current methods choose to downscale the temporal information from coarse images to subpixel scales, which is potentially uncertain, as the fine-resolution spatial information is very limited. In contrast to the current methods, PK-STEM constructs prior FPAR trend surface images with a high spatiotemporal resolution as prior knowledge and uses a simple upscaling linear correction to fuse the new MODIS FPAR images.

Assuming that only systematic bias exists between the prior and real FPAR images on the same date, their relationship can be expressed as

$$F_t = a_t X_t + b_t \tag{4}$$

where F_t and X_t represent the actual and prior Landsat FPAR images, respectively, a_t and b_t are the coefficients of the linear equation, and the subscript t represents date t. Under the assumption, the coefficients a_t and b_t are the same for all Landsat pixels (i.e., global coefficients). Due to the interannual variability of vegetation phenology, the coefficients a_t and b_t can vary with time and are thus derived for each date.

Because the fine FPAR image is retrieved in a scale-consistent manner using high-quality MODIS FPAR values as a reference, the MODIS FPAR can be expressed as a simple average of all subgrid FPAR values from the actual Landsat FPAR image as follows:

$$C_t(u, v) = F_{\mathrm{avg},t}(u, v) = \frac{\sum_{j=1}^{n_F} F_t(j)}{n_F} \tag{5}$$

where C_t is the MODIS FPAR image, $F_{\mathrm{avg},t}$ is the actual Landsat FPAR image aggregated to the MODIS resolution, (u, v) is a given location, $F_t(j)$ represents the jth Landsat pixel, and n_F is the number of Landsat pixels within the given MODIS pixel cell.

Combining (4) and (5), the relationship between the new MODIS FPAR and the prior Landsat FPAR images can be expressed as

$$C_t = a_t X_{\mathrm{avg},t} + b_t \tag{6}$$

where $X_{\mathrm{avg},t}$ represents the prior Landsat FPAR image aggregated to the MODIS resolution. Through (6), we can derive the coefficients a_t and b_t by regressing the MODIS FPAR and aggregated prior Landsat FPAR values on the same date t. The ordinary least squares method is employed to minimize the regression errors. The derived coefficients a_t and b_t are used to correct the systematic bias between the prior and actual Landsat FPAR images on date t as follows:

$$X'_t = a_t X_t + b_t \tag{7}$$

where X'_t is the updated prior Landsat FPAR image by fusing C_t, which is supposed to be more similar to the actual Landsat FPAR image than the un-updated prior image. Combining (2) and (7), the growth in Landsat FPAR from date $t-1$ to t, represented as $\Delta X'_t$, can be expressed as

$$\begin{aligned} \Delta X'_t &= X'_t - X'_{t-1} = (a_t X_t - a_{t-1} X_{t-1}) + (b_t - b_{t-1}) \\ &= (a_t M_t X_{t-1} - a_{t-1} X_{t-1}) + (b_t - b_{t-1}) \\ &= (a_t M_t - a_{t-1}) X_{t-1} + (b_t - b_{t-1}). \end{aligned} \tag{8}$$

Although the coefficients a_t, b_t, a_{t-1}, and b_{t-1} are the same for all Landsat pixels, the growth rates (defined by $a_t M_t - a_{t-1}$) in the updated prior FPAR RSTS data are also determined by the spatially varying prior growth rates (defined by M_t) and thus vary spatially with the Landsat pixels. Similarly, the Landsat FPAR growth ($\Delta X'_t$) determined by X_{t-1} and M_t also maintain spatial variability.

2) Mode II: Fusing New Landsat FPAR Data: PK-STEM employs an EnKS [41] to fuse the new Landsat FPAR data. The EnKS is a sequential data assimilation method within the Kalman gain scheme and uses an ensemble to represent the statistical errors of the model state. It uses an analysis scheme to weigh the errors from predictions and observations and integrates ensemble members to predict the errors of the model state forward in time. Particularly, the EnKS also smooths previous ensemble analysis using present observations.

While the dynamic model in (2) provides short-range FPAR forecasts at 30-m resolution in real time, the EnKS recursively updates the FPAR forecasts as new Landsat observations arrive. The implementation of the EnKS in PK-STEM has four major steps presented as follows.

1) Create an initial ensemble. As FPAR is the only state variable, define a matrix $\mathbf{X} \in \Re^{1\times N}$ containing N ensemble members of the model state. The ensemble mean is $\bar{\mathbf{X}}$, and the ensemble perturbation matrix $\mathbf{P} \in \Re^{1\times N}$ is $\mathbf{P} = \mathbf{X} - \bar{\mathbf{X}}$. Create an initial ensemble $\mathbf{X}_0$ at time $t = 0$ by adding an unbiased Gaussian distributed perturbation

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$\mathbf{P_0}$ to the initial model state, that is, the initial FPAR value from the prior FPAR RSTS data set.

2) Create a forecast ensemble. At time t, compute a forecast ensemble $\mathbf{X}_t$ with the transition model M_t and the analysis ensemble $\mathbf{X}^a_{t-1}$ at $t-1$ using (2), where superscript "a" represents the analysis ensemble.
3) Data assimilation. When new observations arrive, a standard analysis equation is used to create a new analysis ensemble as follows:

$$\mathbf{X}^a_t = \mathbf{X}_t + \mathbf{P}_t\mathbf{P}_t^{\mathrm{T}}\mathbf{H}^{\mathrm{T}}\left(\mathbf{H}\mathbf{P}_t\mathbf{P}_t^{\mathrm{T}}\mathbf{H}^{\mathrm{T}} + \mathbf{R}\right)^{-1}(\mathbf{Y}_t - \mathbf{H}\mathbf{X}_t) \quad (9)$$

where $\mathbf{H} \in \Re^{m\times 1}$ is the measurement operator that relates the model state to observations and m is the number of measurements. $\mathbf{R} \in \Re^{m\times m}$ is the covariance matrix of measurement error, and $\mathbf{Y}_t \in \Re^{m\times N}$ is the perturbed observations ensemble. The superscript "T" represents a matrix transpose.

4) Smooth the analysis backward in time. Given the observation at t, the analysis ensembles of the previous states at time $t'(t' < t)$ are also reanalyzed using (9). Thus, two analysis ensembles of the model state are derived at time t' from forward and backward analysis, respectively. To fully utilize the new observations, we compute a time-weighted combination of the forward and backward analysis as a smoothed analysis using

$$X^s_{t'} = \frac{t_2 - t'}{t_2 - t_1}\bar{\mathbf{X}}^a_{t'(t_1)} + \frac{t' - t_1}{t_2 - t_1}\bar{\mathbf{X}}^a_{t'(t_2)} \quad (10)$$

where $X^s_{t'}$ is the smoothed analysis value at t', t' is the moment between the two available observations at t_1 and t_2 $(t_1 < t_2)$, $\bar{\mathbf{X}}^a_{t'(t_1)}$ is the forward analysis ensemble mean at t' based on the observation at t_1, and $\bar{\mathbf{X}}^a_{t'(t_2)}$ is the backward analysis ensemble mean at t' based on the observation at t_2.

The above introduction to the steps of the EnKS in PK-STEM uses a single Landsat pixel as an example. In practical implementation, the EnKS processes an entire Landsat image as a two-dimensional matrix. All the calculations are performed on matrices and are very efficient.

3) Mode III: Fusing New MODIS and Landsat FPAR Data: PK-STEM can operate in a mode that combines modes I and II, namely, mode III. While mode II directly uses the prior estimate from the dynamic model to initiate ensembles, mode III uses the prior estimate corrected by MODIS FPAR data to initiate ensembles. This combination is meaningful for integrating more spatiotemporal information from both MODIS and Landsat FPAR data.

III. Data and Implementation

A. Study Area and Data

The study area ranges from upstream to downstream of the Heihe River Basin, within 100°8'–100°30' E, 38°28'–39° N (Fig. 2). This area covers approximately 2000 km^2 and features distinct landscapes, including forests, riparian ecosystems, irrigated crops, and deserts, making a detailed representation of vegetation dynamics necessary.

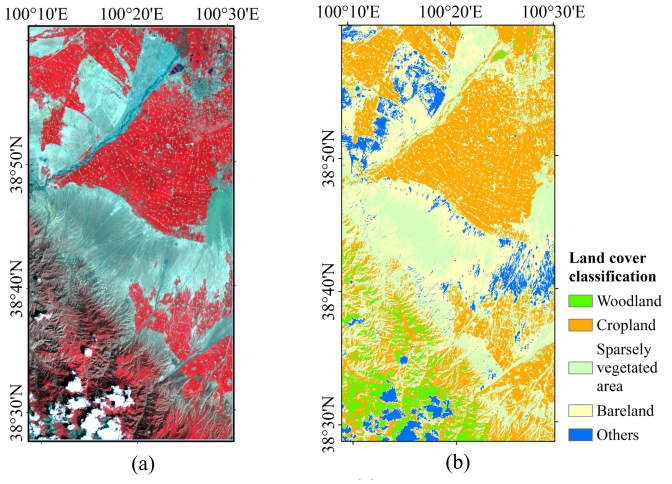


Fig. 2. Map of the study area.

A total of 117 cloud-free Landsat TM\ETM+ images from 2000 to 2012 were collected from Earth Resources Observation and Science (EROS) Center Science Processing Architecture (ESPA, https://espa.cr.usgs.gov/) on-demand interface, which are atmospherically corrected surface reflectance data in UTM/WGS-84 projection. They were classified into five land cover classes, including woodland, cropland, sparsely vegetated area, bare land, and others (e.g., cloud). A previous study in [35] has estimated FPAR from all these 117 Landsat images using the scaling-based method, in which the detailed list of the selected Landsat images can be found. The accuracy of the estimated Landsat FPAR images was validated with the *in situ* measurements, which derives an overall RMSE = 0.072 and mean absolute error (MAE) = 0.058 [35]. The MOD15A2H leaf area index (LAI)/FPAR C6 version products from 2000 to 2012 were downloaded and processed to UTM/WGS-84 projection and 480-m resolution.

B. PK-STEM Implementation

To implement PK-STEM, we composed and filtered the 103 Landsat images from 2000 to 2011 to construct the prior FPAR RSTS data. We carefully analyzed the residual errors of the double logistic fitting. The residual errors of the fitting had an RMSE = 0.08 at the 95% quantile and a mean RMSE = 0.048. To guarantee the temporal continuity of the posterior FPAR RSTS data, we also filtered and smoothed the MODIS FPAR products using the double logistic model. Both the linear correction and EnKS were implemented at 30-m resolution and eight-day intervals (i.e., the same interval as that of MODIS FPAR products) from January 1 to December 31 in a single year. We employed diagonal error covariance matrices for both the dynamic models and observations and set the size of the ensemble to 100, that is, $N = 100$, in the EnKS.

C. Comparison and Evaluation

We designed three experiments to evaluate the performance of PK-STEM: 1) evaluation and comparison of PK-STEM predictions in modes I, II, and III in 2012; 2) comparison

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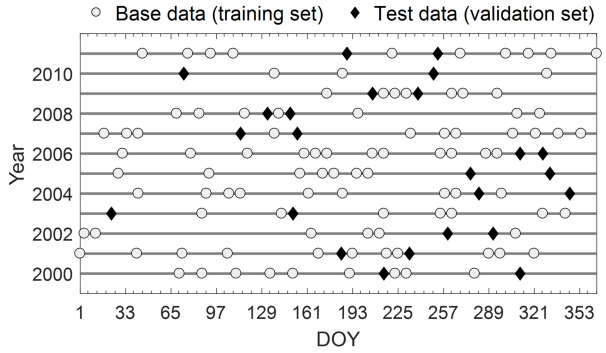


Fig. 3. Temporal distributions of the base and test Landsat FPAR images from 2000 to 2011 as a function of the DOY.

of PK-STEM in mode III with the ESTARFM and LMGM in 2012; and 3) application of PK-STEM in mode III to generate a long time series FPAR data set with 30-m resolution and eight-day intervals from 2000 to 2011.

The performance of PK-STEM in all three experiments was evaluated against the actual Landsat FPAR images. Four statistical indicators were computed between the predicted and actual Landsat FPAR images, including the average difference (AD), RMSE, correlation coefficient (R), and structural similarity (SSIM). AD and RMSE gauged the prediction errors, while R and SSIM [42] were used to evaluate the linear relationship and overall SSIM between the predicted and actual images. The four indicators were computed using the equations as follows:

$$\mathrm{AD} = \mu_X - \mu_F = \frac{\sum_{i=1}^{n}(x_i - f_i)}{n} \tag{11}$$

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}(x_i - f_i)^2}{n-1}} \tag{12}$$

$$R = \frac{\sigma_{FX}}{\sigma_F \sigma_X} = \frac{1}{n-1}\frac{\sum_{i=1}^{n}(f_i - \mu_F)(x_i - \mu_X)}{\sigma_F \sigma_X} \tag{13}$$

$$\mathrm{SSIM} = \frac{(2\mu_F\mu_X + \varphi_1)(2\sigma_{FX} + \varphi_2)}{(\mu_F^2 + \mu_X^2 + \varphi_1)(\sigma_F^2 + \sigma_X^2 + \varphi_2)} \tag{14}$$

where f_i and x_i are pixel values, μ_F and μ_X are the means, σ_F and σ_X are the standard deviations of true and predicted fine images, respectively, n is the total number of pixels in the comparison, σ_{FX} is the covariance of the two images, and φ_1 and φ_2 are two small constants to prevent a null denominator in (14).

In addition, we also compared the time curves at typical pixels from the generated high spatiotemporal resolution FPAR data sets to help evaluate the performance of PK-STEM in capturing the temporal dynamics of vegetation.

In experiments I and II, we divided the fourteen Landsat FPAR images in 2012 into six base images (i.e., training set; corresponding DOY = 32, 112, 176, 224, 272, and 304) and eight test images (i.e., validation set; DOY = 16, 48, 160, 192, 240, 256, 288, and 336) for independent validation. The base Landsat FPAR images were at 50-day intervals on average, corresponding to a common case with limited fine images. In experiment I, we implemented PK-STEM with only 46 MODIS FPAR images in mode I, only six Landsat FPAR images in mode II, and both sets of data in mode III. The PK-STEM predictions in different modes were then evaluated using the same eight test images.

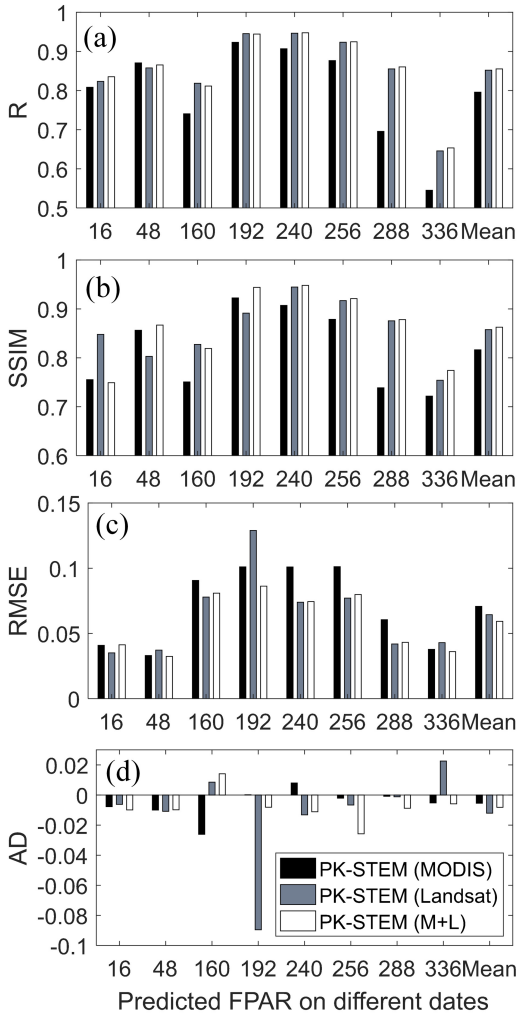


Fig. 4. Accuracy assessment of PK-STEM predictions with only MODIS FPAR (mode I), only Landsat FPAR (mode II) and both data (mode III), against the eight actual Landsat FPAR images in the validation set in 2012.

In experiment II, the performance of PK-STEM in mode III was compared with that of the weight function-based ESTARFM and unmixing-based LMGM methods. The default classification based on thresholds in FPAR was used for the two methods. The other required parameters, such as the search window size and the minimum number of similar pixels in ESTARFM, were all kept as the default values. All three methods used the six training set images as the input base data, and the results were compared with the eight validation set Landsat FPAR images.

In experiment III, PK-STEM in mode III was applied to generate a 30-m resolution FPAR data set at eight-day intervals from 2000 to 2011. We randomly selected two Landsat FPAR images for independent validation and used all the remaining Landsat FPAR images as base images in each year. Fig. 3 shows the temporal distributions of the base and test Landsat FPAR images from 2000 to 2011. The test Landsat FPAR

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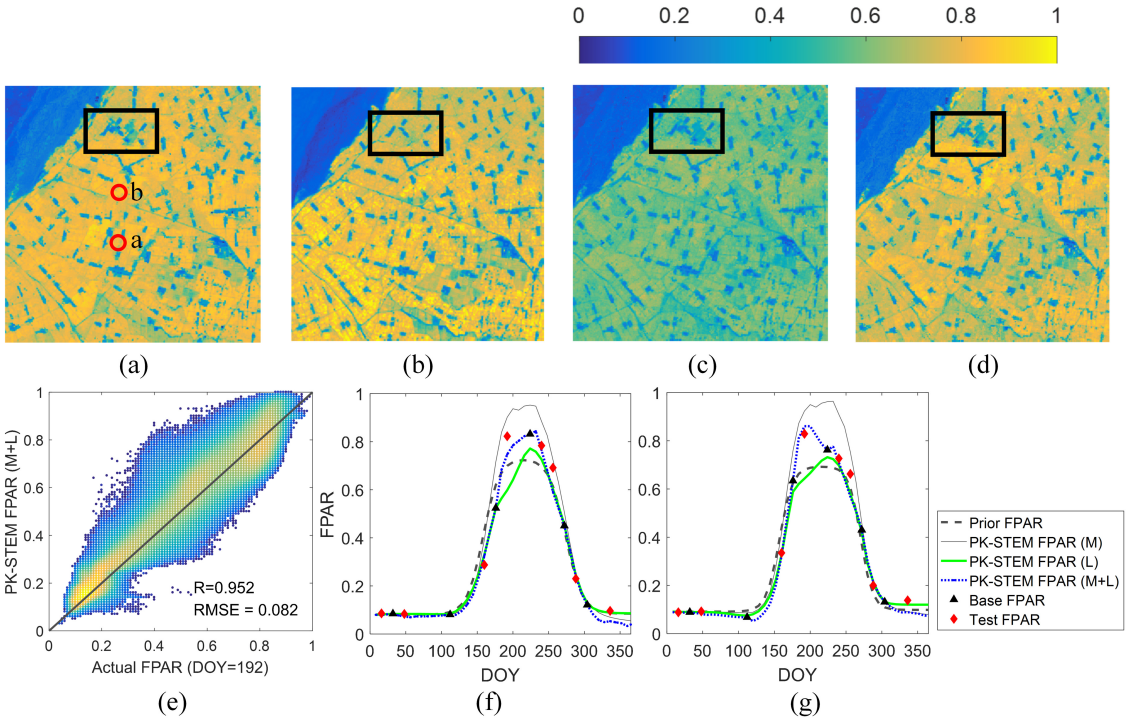


Fig. 5. Comparisons of the (a) actual Landsat FPAR image and FPAR images predicted by PK-STEM in modes (b) I, (c) II, and (d) III in a subarea on July 10 (i.e., DOY 192) in 2012 and the time curves at selected Landsat pixels within these data sets (f and g represent the sites "a" and "b" in a, respectively). (e) Scatter plots between actual and predicted images by PK-STEM in mode III in the entire image on DOY 192. (PK-STEM FPAR (M) = PK-STEM results with only MODIS data, PK-STEM FPAR (L) = PK-STEM results with only Landsat data, PK-STEM FPAR (M+L) = PK-STEM results with both MODIS and Landsat data).

images are evenly distributed across the entire vegetation growth cycles during the 12-year period.

IV. Results

A. PK-STEM Predictions in Different Modes

Fig. 4 compares the quantitative indices computed between all eight actual and predicted images. The PK-STEM results in mode III (i.e., using both MODIS and Landsat FPAR data) have a smaller RMSE and higher R and SSIM values than those in modes I and II in most cases. In general, the PK-STEM results in mode I have a higher RMSE and lower R and SSIM values than those in mode II. However, the PK-STEM prediction in mode II on DOY 192 is an exception, as this prediction has the highest RMSE and a very large negative AD value. This indicates that the available Landsat base data on DOY 176 and 224 do not capture the rapid vegetation growth during this period, resulting in a major underestimate of FPAR on DOY 192. When the new MODIS FPAR data are integrated in mode III, this underestimation on DOY 192 is corrected to a great extent and the prediction of PK-STEM in mode III derives the smallest RMSE value.

For the eight dates, PK-STEM is much more robust in mode III than in modes I and II, with the RMSE ranging from 0.036 to 0.086 (versus 0.033–0.102 and 0.035–0.129), AD ranging from −0.030–0.025 (versus −0.017–0.004 and −0.090–0.023), R ranging from 0.652 to 0.952 (versus 0.556–0.922 and 0.646–0.947), and SSIM ranging from 0.743 to 0.951 (versus 0.717–0.923 and 0.754–0.945, respectively). On average, the accuracies of the three modes of PK-STEM follow the order mode III > II > I, with a mean RMSE = 0.062, 0.064, and 0.071; mean R = 0.851, 0.852, and 0.795; and mean SSIM = 0.863, 0.858, and 0.823, respectively, for the three modes. These quantitative comparisons verified that PK-STEM is more accurate and robust in mode III than in the other two modes during the entire vegetation growth period.

Fig. 5 presents example images on July 10 (i.e., DOY 192) and typical time curves from FPAR RSTS data sets predicted by PK-STEM in different modes in the small study area. In general, all the FPAR RSTS images are spatially continuous with detailed textures. Compared to the actual Landsat FPAR image shown in Fig. 5(a), the result of PK-STEM in mode I presented in Fig. 5(b) shows a slight overestimate in some areas, while that in mode II presented in Fig. 5(c) shows a systematic underestimate. As shown in the black rectangular area in Fig. 5, the results of PK-STEM in modes II and III can reconstruct the correct spatial details, but that in mode I fails to do so. Because PK-STEM in mode I keeps spatial details from the prior FPAR RSTS images, it cannot capture local-scale real-time land cover changes, while PK-STEM in modes II and III can restore such changes from the base Landsat FPAR images. Generally, the FPAR image predicted by PK-STEM in mode III shown in Fig. 5(d) is the most similar to the actual Landsat FPAR image. Fig. 5(e) shows the relationship between the result of PK-STEM in mode III and the actual FPAR image

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TABLE I

ACCURACY ASSESSMENT OF ESTARFM, LMGM, AND PK-STEM AGAINST THE ENTIRE VALIDATION SET OF ACTUAL LANDSAT FPAR IMAGES

	ESTARFM				LMGM				PK-STEM			
DOY	R	SSIM	AD	RMSE	R	SSIM	AD	RMSE	R	SSIM	AD	RMSE
16	0.585	0.595	-0.008	0.051	0.518	0.582	-0.001	0.052	0.845	0.815	-0.013	0.038
48	0.712	0.682	-0.011	0.046	0.642	0.665	-0.009	0.047	0.810	0.845	-0.010	0.036
160	0.748	0.759	-0.026	0.091	0.711	0.723	-0.024	0.094	0.811	0.811	0.021	0.086
192	0.942	0.942	0.005	0.085	0.879	0.878	-0.016	0.121	0.952	0.951	0.025	0.082
240	0.942	0.942	0.012	0.079	0.916	0.917	0.002	0.092	0.943	0.941	-0.022	0.082
256	0.916	0.918	0.007	0.082	0.893	0.895	0.003	0.092	0.927	0.922	-0.030	0.080
288	0.863	0.882	-0.003	0.041	0.817	0.842	-0.004	0.048	0.865	0.871	-0.019	0.049
336	0.502	0.648	0.004	0.048	0.499	0.666	-0.006	0.045	0.652	0.743	-0.012	0.042
Mean	0.776	0.796	-0.002	0.065	0.734	0.771	-0.007	0.074	0.851	0.863	-0.007	0.062

Fill colors from light to dark represent the accuracies from low to high among the three methods.

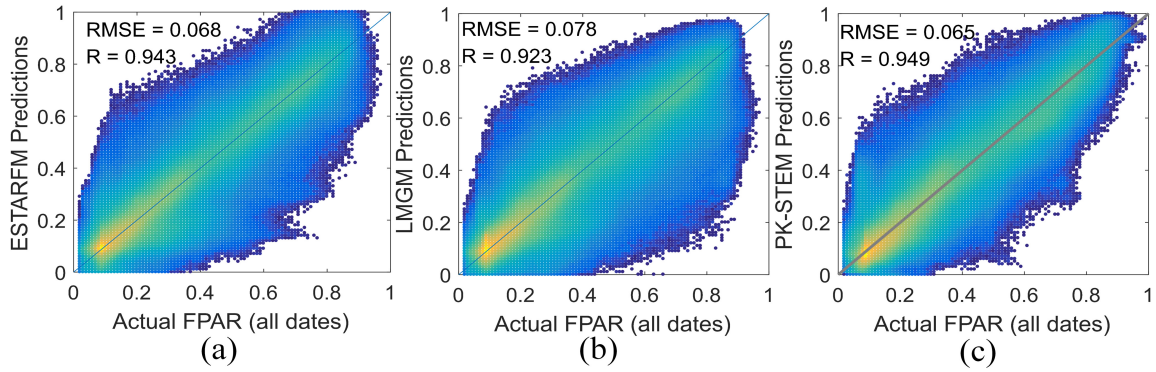


Fig. 6. Scatterplots of actual Landsat FPAR images (x-axis) and the predicted images (y-axis) by (a) ESTARFM, (b) LMGM, and (c) PK-STEM for all eight dates.

in the entire study area, and this relationship verifies the high accuracy of PK-STEM in mode III on DOY 192.

Fig. 5(f) and (g) presents the time curves at typical Landsat pixels within the different FPAR RSTS data sets. The results of PK-STEM in all three modes exhibit smooth time curves. The results of PK-STEM in mode I tend to overestimate FPAR values, while those in mode II show major underestimates, particularly during the rapid vegetation growth period from DOY 176 to 224. The results of PK-STEM in mode III exhibit smooth time curves that are very close to the actual Landsat FPAR values, and these results verify that PK-STEM in mode III can accurately restore vegetation temporal dynamics at a fine resolution.

B. Comparison of the Different Methods

Table I compares the quantitative indices computed between all eight actual and predicted images by the ESTARFM, LMGM, and PK-STEM methods. Clearly, compared with those of the other methods, the PK-STEM results have higher R and SSIM values, lower RMSE values, and larger absolute AD values. Since Landsat FPAR images are highly consistent with MODIS FPAR products and both the ESTARFM and LMGM methods directly weight or unmix the MODIS FPAR pixel values, the deviations of the ESTARFM and LMGM results from the actual Landsat FPAR images may be smaller than those of PK-STEM. However, the AD values in the PK-STEM results range from -0.030 to 0.025, which are very small in an absolute sense.

For the eight dates, PK-STEM is much more robust than the other two methods, particularly in terms of R and SSIM. On average, the accuracies of three methods follow the order PK-STEM $>$ ESTARFM $>$ LMGM, with the mean RMSE $=$ 0.062, 0.065, and 0.074; mean $R = 0.851$, 0.776, and 0.734; and mean SSIM $=$ 0.863, 0.796, and 0.771, respectively, for the three methods. The scatterplots in Fig. 6 compare the actual Landsat FPAR images against the predictions by the three methods on all eight dates, and these results also confirm that the PK-STEM predictions are more accurate than those of the ESTARFM and LMGM.

Fig. 7 compares the actual Landsat FPAR images against the predictions by the ESTARFM, LMGM, and PK-STEM methods in a subarea on DOY 160 and 192 in 2012. Base pairs of MODIS-Landsat images on DOY 112, 176, and 224, as well as the MODIS images on the prediction DOY 160 and 192, are also presented for comparison. From the visual comparison, the FPAR images predicted by all three methods are generally similar to the actual Landsat FPAR

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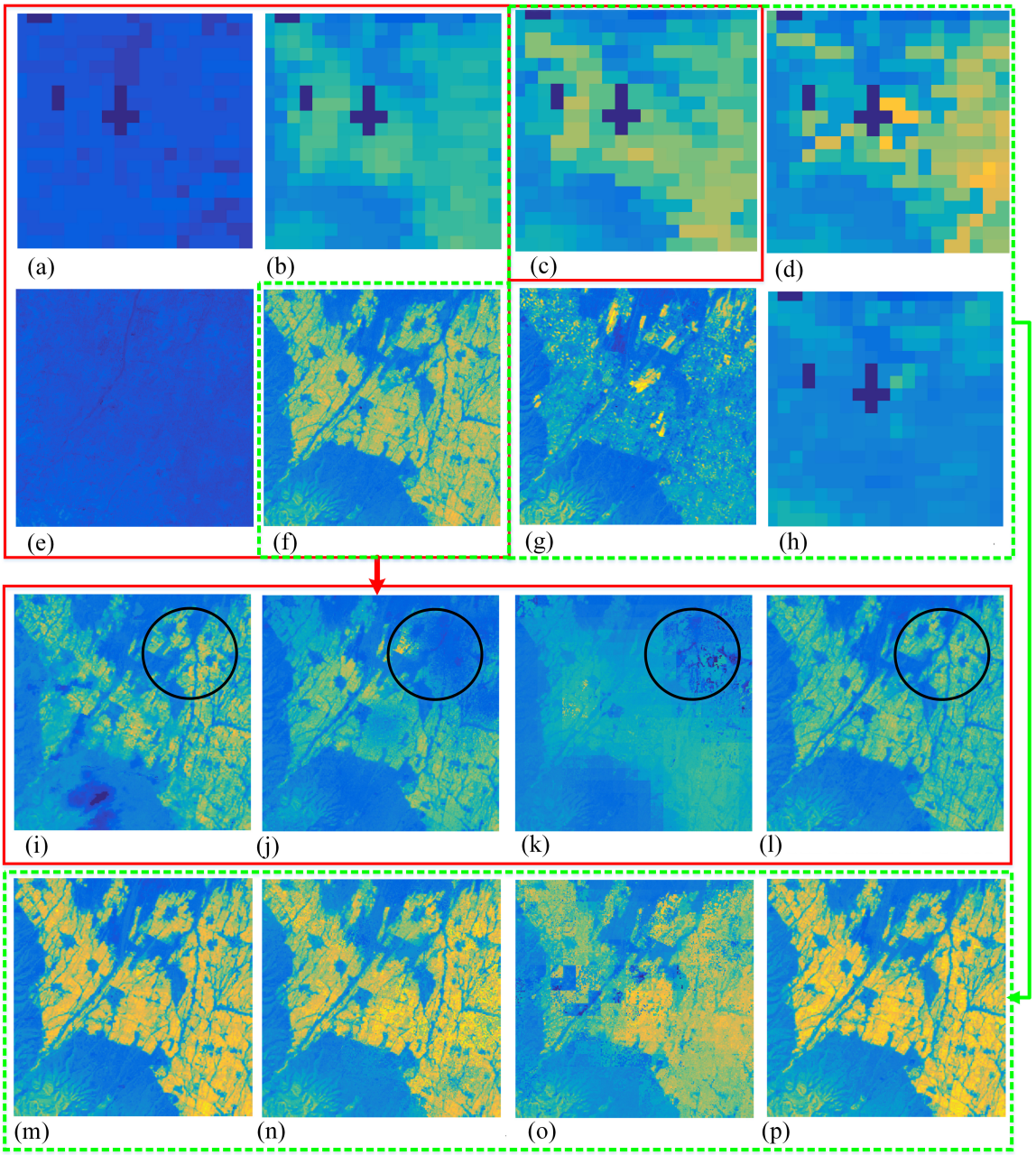


Fig. 7. Comparisons of the actual Landsat FPAR images against the predictions of ESTARFM, LMGM, and PK-STEM on DOY (i)–(l) 160 and (m)–(p) 192. (a)–(h) Base images for predictions on DOY (a)–(c), (e), and (f) 160 and (c), (d), and (f)–(h) 192. (i) and (m) Actual images. (j) and (n) ESTARFM results. (k) and (o) LMGM results. (l) and (p) PK-STEM results.

images, suggesting the capability to capture the temporal changes from the base date to the prediction date. However, on DOY 160, the ESTARFM and LMGM results do not have good spatial continuity in heterogeneous regions and fail to present some spatial details correctly, such as in the black circled area. Because both the ESTARFM and LMGM resolve the rates of linear change for endmembers from the changes at MODIS pixels between the base and prediction dates, they can hardly capture nonlinear temporal changes in the mixed MODIS pixels, as indicated by the obvious underestimation in the black circled area on DOY 160. Both the ESTARFM and LMGM perform better on DOY 192, which may be attributed to the improved spatial homogeneity and steady growth during the peak period from DOY 176 to 224. The LMGM results show significant block effects on DOY 160 and 192. Due to the direct unmixing of neighboring coarse pixels in LMGM, significant block effects inevitably occur when two neighboring coarse pixels have striking contrasts in pixel values. In contrast, PK-STEM performs significantly better than the other two methods, particularly at maintaining better spatial continuity and preserving the shapes of small objects.

In addition to the spatial comparisons, we also compared the pixel-level time curves of the images predicted by the three methods (Fig. 8). In general, all three methods can capture the

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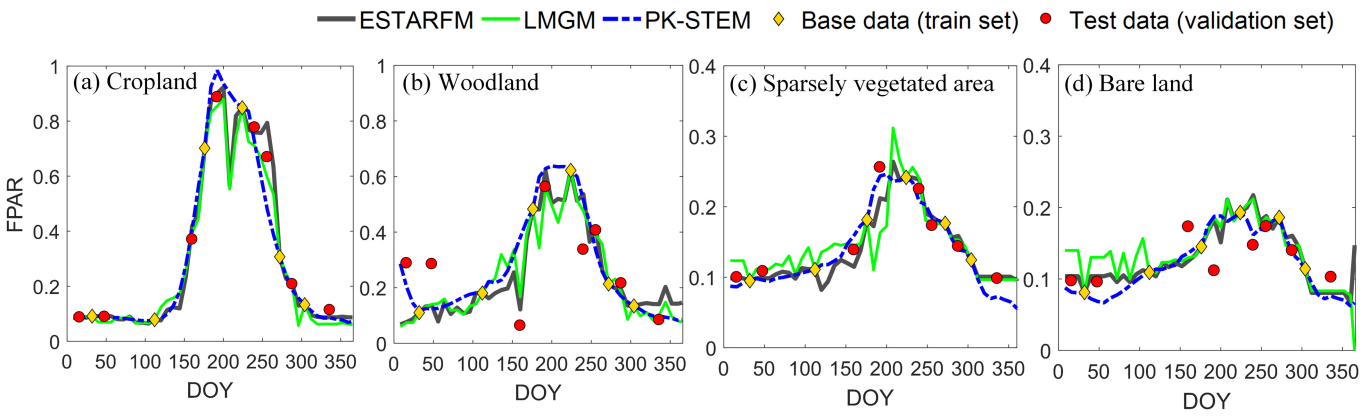


Fig. 8. Comparison of the time curves of the predictions of ESTARFM, LMGM, and PK-STEM at Landsat pixels with different land cover types, including sites of (a) cropland, (b) woodland, (c) sparsely vegetated area, and (d) bare land.

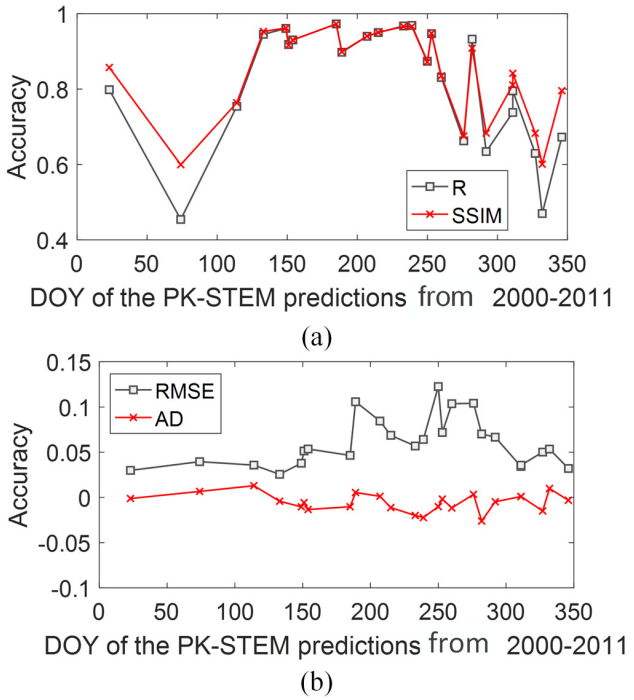


Fig. 9. Accuracy assessment of PK-STEM predictions against the validation set of 24 actual Landsat FPAR images from 2000 to 2011. The accuracy indicators are organized as a function of DOY across different years.

temporal changes for different land cover types. At all these sites, while the ESTARFM and LMGM generate discontinuous time curves with fluctuations, PK-STEM generates smoother time curves that are closer to the actual Landsat observations. Because PK-STEM has better control over the smoothness of the time curves at fine pixels at each step, this method performs significantly better in modeling reliable temporal profiles of vegetation dynamics.

C. Long Time Series of High Spatiotemporal Resolution FPAR Data Set

Fig. 9 shows the quantitative indices computed between all 24 actual images and the images predicted by PK-STEM from 2000 to 2011. The values of four indices clearly exhibit seasonal variations. Both R and SSIM are higher (>0.85), while RMSE (0.026–0.123) and AD (−0.013–0.006) are larger from DOY 149 to 253. Among the predicted images on all dates, 83% of them achieve $R > 0.6$ and RMSE < 0.12, and 92% of them have SSIM > 0.6 and absolute AD ≤ 0.02. On average, the images predicted by PK-STEM are highly accurate, with a mean $R = 0.938$, mean SSIM $= 0.940$, mean RMSE $= 0.066$, and mean AD $= -0.006$. These results indicate that PK-STEM performs very well in predicting long time series of high spatiotemporal resolution FPAR images.

Fig. 10 presents the time curves of PK-STEM predictions and MODIS FPAR products at typical Landsat pixels. Typical Landsat pixels represent each land cover type and are homogeneous at the MODIS resolution. Generally, the time curves of PK-STEM predictions agree well with those of the MODIS FPAR products, indicating that PK-STEM correctly reconstructs the vegetation dynamics at the Landsat resolution. While the MODIS FPAR products exhibit discontinuous time curves with fluctuations, the PK-STEM predictions exhibit smooth time curves and agree with the actual Landsat observations very well. On some dates, even when the Landsat FPAR base values are invalid (i.e., FPAR $= 0$) due to cloud contamination, PK-STEM can correctly reconstruct the time curves at the Landsat resolution, such as in the black circled area in Fig. 10(b). Because PK-STEM employs an EnKS to weigh the prior estimate and new observations, this method can avoid the errors introduced by invalid observations to some extent. However, PK-STEM is still affected by the base Landsat observations. When there are very few base Landsat FPAR values during a rapid vegetation growth period, such as in the black circled area in Fig. 10(d), the performance of PK-STEM may degrade. Generally, both the quantitative evaluations and comparisons of time curves reinforced that PK-STEM is robust and accurate during the entire vegetation growth period and is applicable to long time series applications.

V. Discussion

A. Updates of FPAR Trend Surface Images in Different Modes

PK-STEM incorporates high spatiotemporal information as prior knowledge and updates the prior knowledge by fusing new satellite data in three modes.

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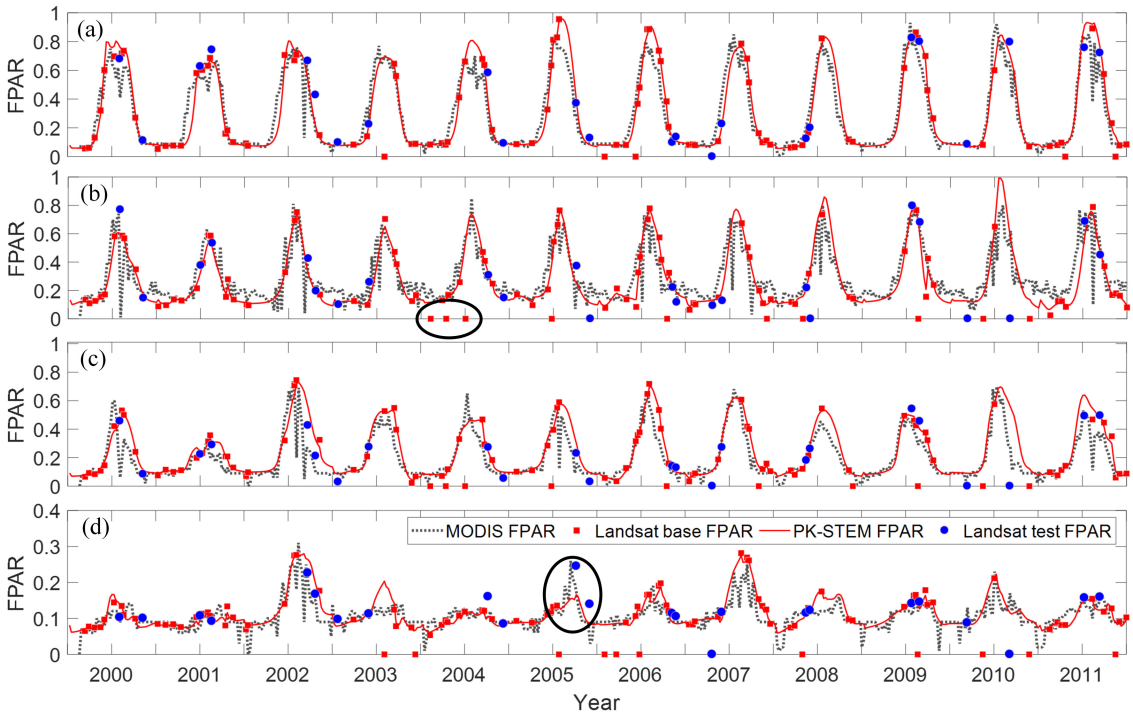


Fig. 10. Time curves of PK-STEM predictions and MODIS FPAR products at typical Landsat pixels with different land cover types, including the sites of (a) cropland, (b) woodland, (c) sparsely vegetated area, and (d) bare land.

Mode I only uses newly observed coarse images to update the prior knowledge without fine images, which might be very useful for extreme cases where clear fine images are not available. Due to the large scale discrepancy between the Landsat and MODIS FPAR images, mode I uses an upscaling linear correction method to correct the systematic bias between the prior RSTS and MODIS FPAR images. A basic assumption is that only systematic bias exists between the prior and actual Landsat FPAR images. This assumption is reasonable when all pixels experience similar changes (i.e., the same linear relationship defined by a_t and b_t) between the prior and actual Landsat FPAR values. Admittedly, changes in land cover types or variations in vegetation growth likely result in small-scale variations in the relationship between the prior and actual FPAR values. However, such small-scale variations could be captured by fusing actual Landsat FPAR images in mode II. In extreme cases with no available Landsat FPAR data, the linear correction is a tradeoff choice because the new information at the Landsat resolution is completely unknown, although this choice might add some error. In addition, the linear correction maintains spatial continuity and variability in the rates and quantities of FPAR growth between consecutive dates, which are inherited from the prior FPAR RSTS data. The advantages of the linear correction in spatial continuity should be attributed to its use of a_t and b_t as global coefficients. An obvious alternative is to compute a_t and b_t for each MODIS pixel, in which spatial discontinuity inevitably occurs between the borders of MODIS pixels with contrasting values. A very similar example relates to the NDVI-downscaling method in [35]. Therefore, the linear correction method is simple and efficient, particularly advantageous for processing large amounts of high spatiotemporal resolution data.

Mode II only uses newly observed fine images to update the prior knowledge. The update is actually a weighting process, depending on whether we place more trust in the prior estimate or new observations. In our experiment, we assigned higher weights for the carefully selected clear new fine images, corresponding to lower variances in the EnKS. These weights are normally equal in the entire image; however, cloud-contaminated or gap pixels are exceptions that we trust only the prior estimate. The output images of such a weighting scheme maintain spatial continuity and can restore the spatial and temporal changes in detail, e.g., changes in land cover and vegetation phenology, which are difficult for mode I. However, mode II is primarily limited to the availability of clear new fine images. Too few clear fine images might result in bias between the prediction and actual images, e.g., significant underestimation might occur without new observations during the vegetation peak period. In cases of very limited new observations, mode III is generally preferred. If clear fine images adequately cover different growth periods for a single year, mode II may be a better choice than modes I and III for its acceptable accuracy and saving the processing of new coarse images. Moreover, mode II using only clear fine images may provide the only efficient tool to generate FPAR products with a high spatiotemporal resolution when coarse images are not available, e.g., before the MODIS era.

Mode III is a combination of modes I and II, which sequentially fuses the new coarse and fine images to update the prior knowledge. Our experiments verified that the predictions of PK-STEM in mode III are more accurate and robust. Fusing

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coarse images maintains the spatiotemporal consistencies of predicted fine images with actual coarse images, while fusing clear fine images restores the spatial and temporal changes in detail. Thus, mode III can work well even when new clear fine images are too sparse in time. Clearly, mode III requires the processing of both coarse and fine images, which might consume more time than modes I and II.

B. Advantages and Limitations of the PK-STEM Method

PK-STEM in mode III outperformed both the ESTARFM and LMGM methods in our experiments with respect to accuracy, spatial continuity, and temporal consistency. PK-STEM has made several improvements over the existing methods. Particular emphasis has been placed on the use of historical fine images and the applicability to long periods. The major strengths of PK-STEM are discussed as follows.

First, PK-STEM reconstructs good spatial continuity and fine spatial details in heterogeneous areas by incorporating prior knowledge with a high spatiotemporal resolution. Weight function-based methods such as ESTARFM use neighborhood information within a moving window for spatially continuous predictions. However, in highly heterogeneous areas where similar pixels are difficult to find, these methods may confront difficulties in reconstructing the correct spatial details. Unmixing-based methods resolve endmember values from neighboring coarse pixels. When two neighboring coarse pixels have striking contrasts in pixel values, spatial discontinuity inevitably appears between fine pixels on the border of neighboring coarse pixels and results in significant block effects in the predicted fine images [25]. As analyzed in the above section, the outputs of PK-STEM are actually linearly corrected prior RSTS images in model I or the weighted-sum of prior and actual fine images in modes II and III. Since all the prior RSTS images are processed from historical actual fine images, PK-STEM is able to maintain good spatial continuity and fine spatial details from actual fine images.

Second, PK-STEM models the temporal profiles at fine pixels in the predicted images well. Both weight function-based and unmixing-based methods directly downscale the temporal information from a coarse pixel to subpixels, which is very uncertain, particularly for highly heterogeneous areas. Unmixing-based methods such as the LMGM also confront major uncertainty in solving overdetermined linear equations. Despite the many strategies developed to constrain endmember values, large uncertainty still exists with the intertwined influences of low imaging and retrieval qualities of coarse images. These uncertainties may appear as fluctuations in the temporal profiles of the fine pixels predicted by weight function-based and unmixing-based methods. To overcome these problems, PK-STEM smooths the new coarse images and employs a bidirectional (i.e., forward and backward) smoothing strategy in fusing new fine images. Therefore, PK-STEM predicts images with better temporal continuity and consistency.

Third, PK-STEM provides a flexible data fusion framework with a continual update mechanism. This method can work well in three different modes by using only new coarse images, only new fine images, or both coarse and fine images. In addition, its update mechanism can use newly available satellite images to enhance the prediction qualities, e.g., satellite data from ASTER, Sentinel-2, and Sentinel-3 sensors. Current spatiotemporal data fusion methods cannot readily accommodate such situations unless reconducting the entire fusion process with the new input data. In contrast, PK-STEM can readily use the previously generated posterior RSTS data as prior knowledge and only perform the following steps, such as the EnKS, for further updates when new data are available.

Finally, PK-STEM has a high computational efficiency, even though it seems to manipulate more data. In PK-STEM, the first step, that is, double logistic fitting at each fine pixel, is the most time-consuming, while the following steps of updating the prior data operate on the entire fine image through matrix calculation and are very efficient. The first step in PK-STEM is similar to that of weight function-based methods to process each fine pixel but does not require neighborhood calculations. For the following updating steps in PK-STEM, our experiments revealed that it only costs approximately 30 s for one-year prediction at eight-day intervals. Thus, PK-STEM may be less efficient than the unmixing-based methods that operate on coarse pixels but is more efficient than ESTARFM for predicting time series images.

In short, the particular advantages with PK-STEM are that it can capture vegetation dynamics more reasonably in terms of spatial details and temporal changes, it provides a flexible data fusion framework with a continual update mechanism, and it is efficient in predicting time series images. Therefore, PK-STEM is particularly suitable and efficient for long-term continuous applications, such as continuous monitoring of crop yield and vegetation phenology over three or more years. Moreover, PK-STEM is an optimal and sometimes may be the only feasible choice when either coarse or fine images are not readily available. For example, in some cloudy areas where clear Landsat images are too limited to cover different vegetation growth periods, existing spatiotemporal fusion methods might confront major problems in time series predictions [43], while PK-STEM with three modes is very flexible to accommodate such situations. Due to this flexibility, PK-STEM has the potential to further extend spatiotemporal fusion to the era of Landsat series satellites (i.e., the 1970s), which is impossible for current spatiotemporal fusion methods that depend on the available MODIS products. Such efforts would be promising to increase our understanding of long-term ecological processes and impacts at a finer scale.

However, the use of historical fine images also leads to the high requirement on prior data, which is a major limitation with PK-STEM. PK-STEM generally needs to compose and process multiyear clear fine images to generate prior RSTS data because clear fine images are always limited due to long revisit periods and unfavorable weather conditions. Although we used all 103 Landsat images from 2000 to 2011 to compose the prior RSTS data in this study, we generally recommend that a minimum of twelve gap-and cloud-free Landsat images that adequately cover the peak season at 20-day intervals or less would be sufficient. Thus, PK-STEM is less ready-to-use for short-term applications (i.e., less than two years). If we do not consider the time for data collection and

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only consider the computation efficiency, we may generally recommend ESTARFM and LMGM in one-off predicting a few fine images, i.e., normally <10, due to their minimal data requirements and acceptable fusion accuracy. Recent developments in big data and cloud computing techniques, such as Google Earth Engine (GEE), would greatly facilitate the manipulation of prior data in PK-STEM. Further efforts can be made to extend PK-STEM to GEE-like platforms, which would further increase the applicability and efficiency of PK-STEM.

Time series filtering techniques on composite RSTS time series data are a major factor affecting the applicability of PK-STEM. Our study area comprises only single-season vegetation and thus the basis form of the double logistic model was used. For double-season vegetation, the double logistic filtering needs to be used in local piecewise form as in [44]–[46] or more flexible filtering techniques can be used, such as Fourier decomposition [44]. PK-STEM is theoretically applicable to other vegetation parameters, such as LAI [47] and vegetation indices (VIs). However, for other land surface parameters indirectly related to vegetation status, such as albedo and land surface temperature (LST), logistic filtering techniques also need to be modified. An alternative solution is to fuse the vegetation parameters (e.g., VI) through the PK-STEM first and then retrieve the nonvegetation parameters (e.g., LST) through the significant relationship between them, such as the relationships between VIs and LST that have been validated in previous studies [48].

The performance of PK-STEM might also be affected by the prediction time intervals. In either the dynamic model or the linear correction method for fusing coarse images, changes in FPAR between consecutive dates are linearly related to the FPAR at the previous date. This is reasonable when time intervals are short, e.g., the eight-day interval in this study. However, when time intervals are long, this linear change assumption may introduce errors due to nonlinear FPAR changes during the rapid vegetation growth period. Since reducing the prediction time intervals will only increase the execution time of the highly efficient EnKS, we generally recommend that prediction time intervals in PK-STEM be no longer than 16-day intervals.

PK-STEM demonstrates a straightforward application of the RSTS framework, yet more applications are expected to embody the profound ideas from the conceptual framework. The vegetation RSTS products by PK-STEM also need to be tested in practical applications, such as crop yield estimation and phenological monitoring, which would benefit a wide range of applications that require vegetation products with a high spatiotemporal resolution.

VI. Conclusion

To improve our capacity to map long-term vegetation dynamics in heterogeneous landscapes, this study proposed a new prior knowledge-based spatiotemporal enhancement method, namely, PK-STEM, to fuse MODIS and Landsat FPAR products following the RSTS framework. PK-STEM integrates ideas from time series filtering, linear correction, and data assimilation into one framework to embody the RSTS methodology. PK-STEM was tested for the Heihe (China) area. We compared the performance of PK-STEM in different modes and with that of two other typical spatiotemporal data fusion methods in 2012. Then, we generated a 13-year FPAR data set at 30-m resolution and eight-day intervals from 2000 to 2012. The main conclusions can be drawn as follows.

1) PK-STEM can work in three different modes by using only MODIS data (mode I), only Landsat data (mode II), or both MODIS and Landsat data (mode III). Our experiments show that mode III derived the highest accuracy with a mean RMSE $= 0.062$ and $R = 0.851$, followed by mode II with a mean RMSE $= 0.064$ and $R = 0.852$ and mode I with a mean RMSE $= 0.071$ and $R = 0.795$. The predictions of PK-STEM in all three modes exhibit good spatiotemporal continuity. However, mode I can hardly capture the changes in spatial details at a fine resolution, while mode II can hardly capture the real temporal changes when the input Landsat images are insufficient during the rapid vegetation growth stage. Mode III combines the strengths of modes I and II and is the most robust in accurately reconstructing the spatiotemporal changes of the landscape at a fine resolution.
2) PK-STEM in mode III (mean RMSE $= 0.062$ and $R = 0.851$) achieves slightly higher accuracies than ESTARFM (mean RMSE $= 0.065$ and $R = 0.776$) on average. The LMGM method derives the lowest accuracies (mean RMSE $= 0.074$ and $R = 0.734$). A particular strength of PK-STEM is in highly heterogeneous areas, where both the ESTARFM and LMGM have difficulty in reconstructing the correct spatial details. PK-STEM maintains more consistent and detailed spatial textures. It also derives smoother and more consistent time curves than the other two methods, indicating its better performance in capturing vegetation growth curves in the predicted images.
3) PK-STEM in mode III achieves a very high accuracy for the 12-year period from 2000 to 2011, with a mean RMSE $= 0.066$ and $R = 0.938$. It derives smooth time curves that agree with the actual Landsat observations very well. In addition, the updating procedure is very efficient and costs only half a minute to predict 46 images at eight-day intervals for an entire year. These conclusions reinforce that the PK-STEM is suitable and efficient for long time series applications.

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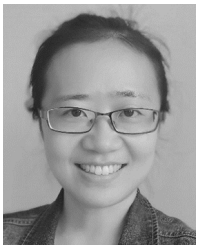
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